MACTFusion: Lightweight Cross Transformer for Adaptive Multimodal Medical Image Fusion

Xinyu Xie, Xiaozhi Zhang, Xinglong Tang, Jiaxi Zhao, Dongping Xiong, Lijun Ouyang, Bin Yang, Hong Zhou, Bingo Wing-Kuen Ling, and Kok Lay Teo

***Abstract*—Multimodal medical image fusion aims to integrate complementary information from different modalities of medical images. Deep learning methods, especially recent vision Transformers, have effectively improved image fusion performance. However, there are limitations for Transformers in image fusion, such as lacks of local feature extraction and cross-modal feature interaction, resulting in insufficient multimodal feature extraction and integration. In addition, the computational cost of Transformers is higher. To address these challenges, in this work, we develop an adaptive cross-modal fusion strategy for unsupervised multimodal medical image fusion. Specifically, we propose a novel lightweight cross Transformer based on cross multi-axis attention mechanism. It includes cross-window attention and cross-grid attention to mine and integrate both local and global interactions of multimodal features. The cross Transformer is further guided by a spatial adaptation fusion module, which allows the model to focus on the most relevant information. Moreover, we design a special feature extraction module that combines multiple gradient residual dense convolutional and Transformer layers to obtain local features from coarse to fine and capture global features. The proposed strategy significantly boosts the fusion performance while minimizing computational costs. Extensive experiments, including clinical brain tumor image fusion, have shown that our model can achieve clearer texture details and better visual quality than other state-of-the-art fusion methods.**

***Index Terms*—Medical image fusion, cross transformer, spatial adaptation fusion, fine-grained information.**

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Xinyu Xie, Xiaozhi Zhang, and Bin Yang are with the School of Electrical Engineering, University of South China, Hengyang 421001, China (e-mail: xxyusc@163.com; zxz_usc@163.com; yangbin01420@163.com).

Xinglong Tang, Jiaxi Zhao, Dongping Xiong, and Lijun Ouyang are with the School of Computing/Software, University of South China, Hengyang 421001, China (e-mail: x2783599597@163.com; m18231192630@163.com; xiongdongping@163.com; layjung@163.com).

Hong Zhou is with the Department of Radiology, The First Affiliated Hospital, University of South China, Hengyang 421001, China (e-mail: zhouhong@msn.cn).

Bingo Wing-Kuen Ling is with the School of Information Engineering, Guangdong University of Technology, Guangzhou 510006, China (e-mail: yongquanling@gdut.edu.cn).

Kok Lay Teo is with the School of Mathematical Science, Sunway University, Kuala Lumpur 47500, Malaysia, and also with the School of Electrical Engineering, Computing and Mathematical Sciences, Curtin University, Perth, WA 6845, Australia (e-mail: k.l.teo@curtin.edu.au).

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I. Introduction

MULTIMODAL medical imaging plays a vital role in disease diagnosis and provides valuable assistance to medical practitioners. According to imaging mechanisms, medical images are usually divided into two categories, i.e., structural and functional information images [1]. The structural images have high spatial resolution and can depict anatomical structures of organs. For instance, Computed Tomography (CT) can display high-density structures like bones, and Magnetic Resonance Imaging (MRI) provides valuable information on soft tissues like adipose and muscular tissue [2]. However, CT images lack detailed information on structural organization, and MRI images lack high-contrast information. Functional images such as Positron Emission Tomography (PET) and Single Positron Emission Tomography (SPECT), can reflect the blood flow and metabolic activity of biomolecules. However, the spatial resolution of functional images is low, which limits the sharpness and detailed descriptions. The purpose of multimodal image fusion is to fuse complementary information from different images to enhance visual perception [3], [4]. Illustrations of CT/PET/SPECT images, MRI images, and fused results are shown in Fig. 1.

In recent years, various medical image fusion methods have been developed. These methods can be divided into two categories, namely classical methods and deep learning methods. The classical fusion methods include multiscale transform (MST) [5], [6], [7], pulse coupled neural network (PCNN) [8], [9], and sparse representation [10], [11]. For instance, shearlet [5], discrete wavelet [6], and low-rank representation [7] have been successfully applied in image fusion. Pulse coupled neural network is used to design fusion rules, but its performance is limited by parameter selection [8]. Sparse representation (SR) simulates the sparse coding principle of visual system in image fusion [10], but dictionary learning is time-consuming. Although these classical methods have demonstrated favorable performance, they still face drawbacks such as irreversible data loss and detail distortion [12]. In addition, the design of fusion rules is laborious and complex.

Recently, deep learning has been employed to address the aforementioned limitations [13]. These deep learning methods mainly include convolutional neural networks (CNNs) [14],

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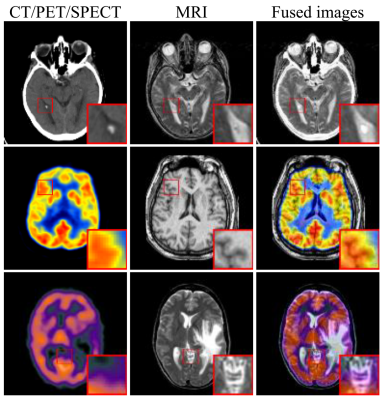


Fig. 1. Illustrations of multimodal medical image fusion. From left to right: CT/PET/SPECT images, MRI images, and fused images.

[15], [16], [17], [18], [19], generative adversarial networks (GANs) [20], [21], and Transformers [22], [23]. Generally, the fusion process consists of three parts, namely feature extraction, feature fusion strategy and image reconstruction [14]. The key to the fusion performance lies in the first two parts [16]. Although deep learning methods have shown advantages and addressed some shortcomings of traditional algorithms, there are still several challenges that need to be addressed. In particular, CNN-based frameworks capture local features through convolution operations while ignoring long-range interactions, resulting in the loss of global contextual features.

The recent Transformers have overcome the limitations of global feature extraction, and it is initially used in natural language processing (NLP) [24]. It has achieved excellent results in image processing such as image recognition [25] and image segmentation [26], [27]. Transformers have also promoted the development of image fusion [22], [23]. However, current Transformer based fusion methods still have limitations. Firstly, most networks adopt simple fusion strategies such as concatenation or addition, resulting in low efficiency in utilizing multimodal image information. These networks usually mine intra-domain interactions, but cannot fully mine and integrate the inter-domain information from multimodal images. Moreover, the problem of inconsistent data distribution in multimodal images has not been effectively solved, and direct cross-fusion operations may cause information interference and redundancy. In addition, Transformer-based methods are computationally expensive. Due to the computation of multi-head attention and self-attention, it needs a high computational cost and long running time. Therefore, it is necessary to explore the combination of local and global methods to enhance flexibility. Recently, MaxVit introduced a multi-axis attention mechanism, attempting to learn interactions between global and local spaces, and has achieved satisfactory results [28].

Inspired by the successful application of vision Transformer, in this work, we develop an adaptive cross-modality fusion strategy for unsupervised medical image fusion. The proposed cross-modality fusion Transformer consists of cross multi-axis attention, which includes cross-window attention and cross-grid attention. It aims to mine and integrate both local and global interactions of multimodal features at lower computational costs. Moreover, the cross Transformer is guided by a novel spatial adaptation fusion module that can adaptively remap feature distributions. This design allows to learn rich textures and appropriate intensity information. The main contributions of this work are summarized as follows.

- We propose an adaptive cross-modality fusion framework for unsupervised multimodal medical image fusion. It can effectively mine and integrate meaningful information, while improving fusion performance and reducing computational costs.
- We design a novel lightweight cross Transformer with a multi-axis attention mechanism. It includes cross-window attention and cross-grid attention, aiming to integrate local and global interactions of multimodal features.
- We introduce a novel spatial adaptation fusion module to adaptively remap feature distributions. It guides the cross Transformer to the most relevant information.
- We design a special feature extraction module that combines multiple residual dense gradient convolutional and Transformer layers to obtain local features from coarse to fine, and effectively capture global features.

In Section II, we review the relevant literatures. Section III discusses the proposed framework, emphasizing the key design. Section IV presents extensive qualitative and quantitative experiments. In Section V, the conclusion and future research are discussed.

II. Related Work

A. Deep Learning Image Fusion

Deep learning based fusion methods significantly improve the fusion performance due to its powerful representation capability. Liu et al. developed a CNN-based method to generate fusion weight maps via multiscale pyramid approach [13]. However, the fusion strategy requires complex manual design. In another work, Liu et al. proposed a supervised framework trained on high-quality images to solve multi-focus image fusion [14]. Zhang et al. developed a supervised network including feature extraction, feature fusion, and image reconstruction [15]. However, these methods do not fully exploit the characteristics of images, and there are some mosaics in functional images that may blur the fusion results. Furthermore, training such an end-to-end model without the ground truth is a challenge. Zhang et al. further designed a unified image fusion model that uses a special loss function to preserve the intensity and gradient information [17]. They subsequently proposed a squeeze-and-decomposition framework to improve the fidelity [18]. Although these models have obtained relatively satisfactory fusion images, the structural information may be lost due to only considering intensity and gradient information [19]. Ma et al. developed a GAN-based model (FusionGAN) [20], which fed source images to a generator and fusion images were obtained by a discriminator. Later, they introduced a dual-discriminator conditional GAN model (DDcGAN) [21], which employed two discriminators to distinguish structure differences between fused images and source images. However, the training process is unstable and texture details are distorted.

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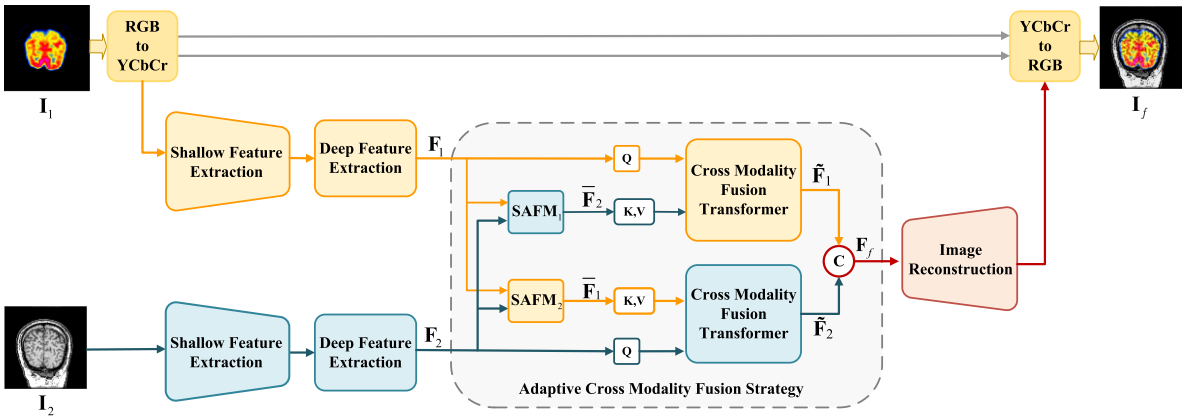


Fig. 2. Overall framework of the proposed MACTFusion for PET and MRI image fusion.

CNN-based image fusion methods still have limitations. They fail to capture global contextual information in the multimodal images. In addition, most of them still rely on traditional manual fusion strategies, which limit the fusion performance.

B. Vision Transformer

Transformer has achieved significant success in natural language processing. Due to its superior performance in global feature extraction, researchers are exploring Transformer-based models in computer vision, including visual recognition [25], object detection [26], and image segmentation [27]. Several related works have achieved excellent results in image fusion [33], [34]. However, they mainly focus on exploring intra-domain interactions and cannot fully integrate inter-domain contexts. More recently, SwinFusion introduced a cross Transformer model in the fusion strategy, which has achieved superior fusion performance [1]. Its computational cost is also high [29], as obtaining global interactions requires a considerable amount of computation. Therefore, effectively integrating global and local interactions remains a challenge. Tu et al. proposed a Transformer based on multi-axis attention mechanism, called MaxViT, aimed to learn local and global interactions [28]. It has achieved state-of-the-art (SOTA) results on various tasks including image generation, object detection, and image classification.

Inspired by the above discussions, we design an adaptive cross Transformer fusion strategy that aims to effectively integrate the intra-modal and inter-modal interactions. Additionally, we also take into account the computational cost. This provides a promising direction for future image fusion.

III. Methodology

A. Overall Framework

In this work, we mainly focus on the fusion of MRI with other medical images, such as CT, PET and SPECT images. In CT-MRI fusion, we hope to preserve texture information of MRI images while retaining intensity information of CT images. MRI can be directly fused with CT. In PET-MRI and SPECT-MRI fusion, we expect the functional information (color information) in PET/SPECT images to be maintained. PET and SPECT images first need to be converted from RGB space to YCbCr space. This conversion provides brightness information of PET and SPECT images to channel Y, which is then fused with MRI. The fused Y channel further combines with Cb and Cr channels. Finally, the YCbCr space is converted back to RGB space again to generate the fused image. It is worth noting that there are similar operations for other biomedical image fusion tasks in this work. Generally, we expect fused images to obtain rich textures and appropriate intensity information. Fig. 2 shows the overall framework for MRI-PET fusion using the proposed MACTFusion. $\mathbf{I}_1 \in \mathbb{R}^{H\times W\times C}$ and $\mathbf{I}_2 \in \mathbb{R}^{H\times W\times C}$ represent two aligned medical images with different modalities respectively, where W, H and C denote the width, height, and number of channels. MACTFusion aims to integrate meaningful information to obtain the fusion image $\mathbf{I}_f$. It consists of three parts, namely feature extraction, adaptive cross-modality fusion and image reconstruction.

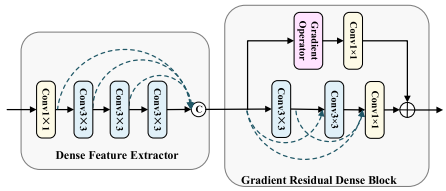


Fig. 3. Architecture of shallow feature extraction module.

1) Feature Extraction: Feature extraction includes shallow and deep feature extraction. The shallow feature extraction consists of a dense feature extractor and a gradient dense block (GRDB). In Fig. 3, in dense feature extractor, a 1×1 convolutional operation is used to reduce the channel dimension. The following three cascaded 3×3 convolution layers are used to extract local features, and their outputs are concatenated together. Skip connections between these convolution layers help to deepen the number of layers. Then the output is fed to GRDB that can enhance the learning of fine-grained details through gradient operations [30]. In GRDB, the mainstream deploys two 3×3 convolution layers and one 1×1 convolution

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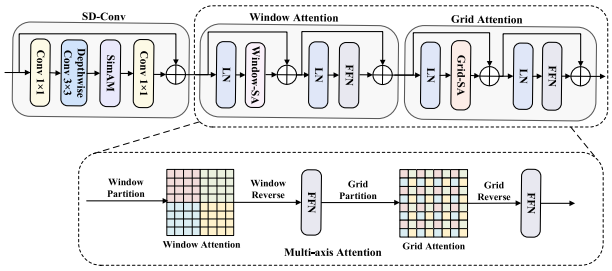


Fig. 4. Network structure of deep feature extraction module.

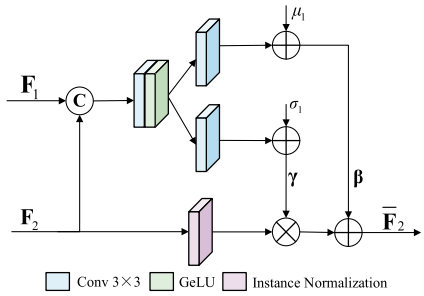


Fig. 5. Architecture of SAFM_1.

layer with dense connections. The residual stream employs the gradient operation to compute gradient magnitude of features, followed by a 1×1 convolution layer to address the difference in channel dimensions. The outputs of two streams are combined together via element-wise addition.

In Fig. 4, it illustrates the network structure of deep feature extraction module, which consists of a simple depth-wise separable convolution (SD-Conv) block and a multi-axis attention block. The designed SD-Conv is constructed with two 1×1 convolution layers, a depth-wise 3×3 convolution layer, and a simple parameter-free attention module (SimAM) [31], [32]. It is worth noting that using SD-Conv together with attention can further improve the generalization and trainability of the model [28]. Moreover, placing the SD-Conv block before multi-axis attention provides another advantage. SD-Conv can be regarded as positional encoding generator [33], which makes the model free of explicit positional encoding layers.

The proposed multi-axis attention is decomposed into two sparse parts, i.e., window attention and grid attention. It can reduce the computational complexity from quadratic to linear. Let the size of input feature be $H \times W \times C$, where H, W and C are the height, width and channel number, respectively. As shown in Fig. 4, the window attention reshapes the input feature to size $(\frac{H}{P} \times \frac{W}{P},\ P \times P,\ C)$ by partitioning it into non-overlapping blocks, where the size of each block is $P \times P$. Then self-attention is performed on these local blocks rather than on the entire feature map, with the aim of performing local interactions. Furthermore, we introduce grid attention to perform global interactions. It partitions feature maps into grid sizes. Then self-attention is performed along the grid axis, corresponding to global attention. By stacking the window attention and grid attention to form multi-axis attention blocks, both local and global interactions can be obtained. With the fixed block and grid size, the computational cost is reduced to linear complexity. The attention mechanism can be defined as

$$\text{Attention}\left(\mathbf{Q},\mathbf{K},\mathbf{V}\right) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathrm{T}}}{\sqrt{d}} + \mathbf{B}\right)\mathbf{V} \qquad (1)$$

where $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ represent the query matrix, key matrix and value matrix, respectively. d is the key dimension and $\mathbf{B}$ is the relative positional encoding [1]. As aforementioned, the positional encoding $\mathbf{B}$ of input feature can be generated through SD-Conv [33]. The multi-axis attention block is also equipped with the feedforward network, LayerNorm and skip connections. We stack SD-Conv blocks and multi-axis attention blocks sequentially to extract deep features.

2) Adaptive Cross-Modality Fusion Strategy: We further design a novel adaptive cross-modality fusion strategy to integrate features and generate fused feature $\mathbf{F}_f$. In Fig. 2, the proposed cross-modality fusion strategy consists of two parallel symmetric branches, and each branch includes a spatial adaptation fusion module (SAFM) and a cross-modality fusion Transformer. It handles features as queries, keys and values, and exchanges information between the two branches, aiming to integrate more complementary information from different medical images.

Multimodal medical image pairs usually contain latent structural similarities, and each medical image has its own specific attributes. And their colour and brightness distributions are inconsistent, which can easily cause interference during fusion [29]. Therefore, simply concatenating image features together is not optimal. Inspired by spatially adaptive normalization [34], [35], [36], we introduce a new spatial adaptation fusion module (SAFM) to remap the feature distribution of images. Fig. 5 illustrates the architecture of SAFM_1 in the upper branch of cross-modality fusion strategy (in Fig. 2, SAFM_2 has a similar structure). In Fig. 5, $\mathbf{F}_2$ is modulated by parameters $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ to remap its distribution to that of $\mathbf{F}_1$. $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ are initialized to zero, which can be calculated through concatenation and convolution operations of $\mathbf{F}_1$ and $\mathbf{F}_2$. Let channel-wise mean and standard deviation of $\mathbf{F}_1$ be μ_1 and σ_1, respectively. We update $\boldsymbol{\beta}$ as $\boldsymbol{\beta} \leftarrow \boldsymbol{\beta} + \mu_1$ and $\boldsymbol{\gamma}$ as $\boldsymbol{\gamma} \leftarrow \boldsymbol{\gamma} + \sigma_1$. In addition, we apply point-wise instance normalization to $\mathbf{F}_2$ as

$$\mathbf{F}_2 \leftarrow \frac{\mathbf{F}_2 - \mu_2}{\sigma_2} \qquad (2)$$

where μ_2 is the mean, and σ_2 is the standard deviation of $\mathbf{F}_2$. Then, the normalized $\mathbf{F}_2$ is multiplied with $\boldsymbol{\gamma}$, and their product is further added with $\boldsymbol{\beta}$ to obtain output $\bar{\mathbf{F}}_2$, that is $\bar{\mathbf{F}}_2 = \mathbf{F}_2 \odot \boldsymbol{\gamma} + \boldsymbol{\beta}$, where $\odot$ denote elementwise multiplication. During training process, $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ are updated at each iteration. Thus, SAFM_1 helps $\mathbf{F}_2$ to approximate the distribution of $\mathbf{F}_1$. Similarly, we can obtain $\bar{\mathbf{F}}_1$ via SAFM_2 in the lower branch of cross-modality fusion strategy.

It is worth noting that in the cross-modality fusion Transformer, input features of SD-Conv are modulated by SAFM to generate spatially adaptive modulation feature maps. Due to that SD-Conv generates the positional encoding of these feature maps for attention mechanism, SAFM can refine the positional

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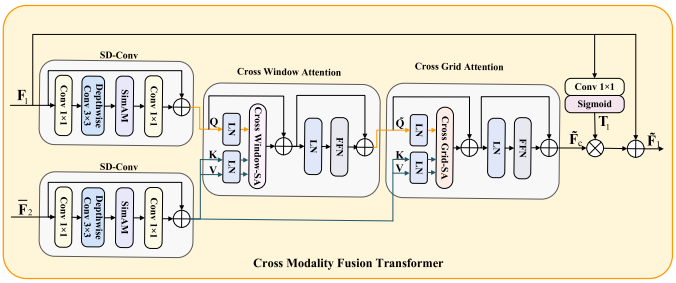


Fig. 6. Architecture of cross-modality fusion transformer.

coding and guide the cross Transformer to focus on the most relevant information and enhance the fusion performance.

Following the spatial adaption fusion module, we design a novel cross-modality fusion Transformer to further mine and integrate local interactions and global interactions of multimodal features. Fig. 6 illustrates the architecture of cross-modality fusion Transformer in the upper branch of cross-modality fusion strategy (in Fig. 2). It consists of SD-Conv blocks and cross multi-axis attention. The cross multi-axis attention is an improved multi-axis attention, which includes cross-window attention and cross-grid attention. As shown in Fig. 6, the features $\mathbf{F}_1$ and $\bar{\mathbf{F}}_2$ are first processed via two SD-Conv blocks respectively, and their outputs are treated as the query ($\mathbf{Q}$), key ($\mathbf{K}$) and value ($\mathbf{V}$) feature accordingly.

These features are fed into the cross-window attention to perform local interactions and obtain the output query ($\tilde{\mathbf{Q}}$). Subsequently, we feed the obtained query ($\tilde{\mathbf{Q}}$) and the same key ($\mathbf{K}$), value ($\mathbf{V}$) to the cross-grid attentionss to perform global interactions. The cross-window attention and cross-grid attention are also equipped with feedforward network (FNN) [37], LayerNorm (LN) and skip connections. The procedure can be represented as

$$\begin{aligned}
\mathbf{F}_{\mathrm{c}} &= \text{CrossWindow Attention}\left(\mathrm{LN}\left\{\mathbf{Q},\mathbf{K},\mathbf{V}\right\}\right)+\mathbf{Q},\\
\tilde{\mathbf{Q}} &= \mathrm{FFN}\left(\mathrm{LN}\left(\mathbf{F}_{\mathrm{c}}\right)\right)+\mathbf{F}_{\mathrm{c}},\\
\bar{\mathbf{F}}_{\mathrm{c}} &= \text{CrossGridAttention}\left(\mathrm{LN}\left\{\tilde{\mathbf{Q}},\mathbf{K},\mathbf{V}\right\}\right)+\tilde{\mathbf{Q}},\\
\tilde{\mathbf{F}}_{\mathrm{c}} &= \mathrm{FFN}\left(\mathrm{LN}\left(\bar{\mathbf{F}}_{\mathrm{c}}\right)\right)+\bar{\mathbf{F}}_{\mathrm{c}}
\end{aligned} \tag{3}$$

where $\tilde{\mathbf{F}}_{\mathrm{c}}$ is the output of cross-grid attention. Then, the output of cross-modality fusion Transformer can be expressed as

$$\tilde{\mathbf{F}}_1 = \tilde{\mathbf{F}}_{\mathrm{c}} \otimes \mathbf{T}_1 + \mathbf{F}_1 \tag{4}$$

where $\mathbf{T}_1$ is the output of sigmoid function with $\mathbf{F}_1$ as the input. The proposed cross-modality fusion Transformer absorbs features from $\bar{\mathbf{F}}_2$ that contribute to $\mathbf{F}_1$. Similarly, we can obtain $\tilde{\mathbf{F}}_2$ via the symmetric cross-modality fusion Transformer in the lower branch of cross fusion strategy. Finally, we concatenate the two output features $\tilde{\mathbf{F}}_1$ and $\tilde{\mathbf{F}}_2$ to obtain the final fused feature $\mathbf{F}_f$. Here, we use concatenation instead of the previous SAFM for fusion, due to that SAFM is mainly used to adaptively remap feature distributions for the following cross Transformer. In addition, adopting concatenation as the final fusion step is simple and effective [1].

3) Image Reconstruction: We reconstruct the fusion image $\mathbf{I}_f$ from the fused feature $\mathbf{F}_f$ by image reconstruction module. It can be presented as

$$\mathbf{I}_f = H_{\mathrm{R}}\left(\mathbf{F}_f\right) \tag{5}$$

where $H_{\mathrm{R}}(\cdot)$ is the image reconstruction operator. Image reconstruction contains three convolution layers with LReLU activation function. They are one 1×1 convolution layer and two 3×3 convolution layers.

B. Loss Function

We consider the loss function from several aspects. On the one hand, we define a structure similarity index measure (SSIM) loss and a gradient loss (texture loss) for guaranteeing the visual fidelity. On the other hand, we define an intensity loss to constrain the fusion module to obtain appropriate intensity information.

1) SSIM Loss: SSIM can measure the degree of distortion and the degree of similarity between two images [38]. We use SSIM loss to guarantee structure similarity between fused images and source images. It can be defined as

$$\begin{aligned}
\mathcal{L}_{\mathrm{ssim}} = {} & \eta\cdot\left(1-\mathrm{SSIM}\left(\mathbf{I}_f,\mathbf{I}_1\right)\right)\\
& +\left(1-\eta\right)\cdot\left(1-\mathrm{SSIM}\left(\mathbf{I}_f,\mathbf{I}_2\right)\right)
\end{aligned} \tag{6}$$

where $\mathrm{SSIM}(\cdot)$ is used to calculate the similarity between two images. Weight η is used to control the degree of structure preservation. Since both source images are equally important, the value of η is set to 0.5.

2) Gradient Loss: Integrating more texture details is essential to improve visual quality. We use gradient loss to ensure that more fine-grained details are preserved, which is defined as

$$\mathcal{L}_{\mathrm{grad}} = \frac{1}{HW}\left\|\left|\nabla\mathbf{I}_f\right|-\max\left(\left|\nabla\mathbf{I}_1\right|,\left|\nabla\mathbf{I}_2\right|\right)\right\|_1 \tag{7}$$

where $\|\cdot\|_1$ and $|\cdot|$ represent the $l_1-\mathrm{norm}$ and absolute operator, respectively. ∇ denotes the Sobel gradient operator. $\max(\cdot)$ represents the maximum selection of element-wise.

3) Intensity Loss: Generally, there is significant intensity information and contrast information in CT and PET/SPECT images. We employ intensity loss to ensure that appropriate intensity information is retained. It can be defined as

$$\mathcal{L}_{\mathrm{int}} = \frac{1}{HW}\left\|\mathbf{I}_f-\max\left(\mathbf{I}_1,\mathbf{I}_2\right)\right\|_1 \tag{8}$$

where $\max(\cdot)$ is the element-wise maximum selection operation, and $\|\cdot\|_1$ denotes the l_1-norm.

Finally, we join the structural similarity loss $\mathcal{L}_{\mathrm{ssim}}$, gradient loss $\mathcal{L}_{\mathrm{grad}}$ and intensity loss $\mathcal{L}_{\mathrm{int}}$ as the total loss as

$$\mathcal{L} = \delta_1\mathcal{L}_{\mathrm{ssim}} + \delta_2\mathcal{L}_{\mathrm{grad}} + \delta_3\mathcal{L}_{\mathrm{int}} \tag{9}$$

where δ_1, δ_2, and δ_3 are the weights to control the trade-off between $\mathcal{L}_{\mathrm{ssim}}$, $\mathcal{L}_{\mathrm{grad}}$ and $\mathcal{L}_{\mathrm{int}}$.

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IV. Experiments and Discussions

A. Comparison Methods and Evaluation Metrics

Comparison Methods: We perform extensive comparison experiments with state-of-the-art methods to demonstrate the validity of MACTFusion. These comparison methods include CSMCA [39], EMFusion [40], FusionGAN [20], IFCNN [15], U2Fusion [16], IFT [22] and SwinFusion [1]. CSMCA is the traditional fusion method. EMFusion, FusionGAN, IFCNN and U2Fusion are CNN-based fusion methods. IFT and SwinFusion are Transformer-based structures. We use the public codes with corresponding parameter settings.

Evaluation Metrics: We carry out the qualitative and quantitative evaluation of fusion performance. Qualitative evaluation is an intuitive evaluation based on visual perception. Quantitative evaluation is an objective assessment based on quantitative metrics. Six common quantitative metrics are chosen including sum of correlation differences (SCD) [41], multiscale structure similarity index measure (MS-SSIM) [42], edge information measurement ($Q^{AB/F}$) [43], standard deviation (SD) [43], weighted fusion quality metric (Q_w) [44], and visual information fidelity (VIF) [45]. SCD reflects the transferred complementary information. MS-SSIM measures the structure similarity between fused images and source images. $Q^{AB/F}$ is the gradient metric that can evaluate the retention degree of structure information and edge information. The higher $Q^{AB/F}$, the better fusion effect. SD reflects the contrast and distribution of grayscale values with statistics. The higher SD, the higher contrast, and better visual impact. Q_w reflects the salience information. The higher Q_w, that more saliency information is retained, corresponding to important perceptual region. VIF calculates the visual fidelity between the fusion images and source images. In experiments, the batch size is 4 and, Adam optimizer with a learning rate 0.0001 is used [28]. The experiments are conducted using Nvidia GeForce RTX 3090 GPU and 3.60 GHz Intel Core i9-9900K CPU.

B. CT-MRI, PET-MRI and SPECT-MRI Image Fusion

In this section, we conduct experiments on three typical fusion tasks, namely CT-MRI, PET-MRI, and SPECT-MRI. We use Harvard Medical Dataset as the training and test dataset independently. There are 166 CT-MRI image pairs, 329 PET-MRI image pairs and 539 SPECT-MRI image pairs. The size of images is 256×256. We augment image pairs to 30000 via image rotation [46], which can improve the model and help to combat potential overfitting. 21 pairs of test images are randomly chosen to evaluate the model. In CT-MRI image fusion, since both source images are grayscale, they can be directly fused. In PET-MRI and SPECT-MRI image fusion, since PET and SPECT images contain color information, we need to convert them form RGB space to YCbCr space using (10) [40]. The Y channel represents the brightness variation and structural details, which is used to fuse with MRI images.

$$\begin{pmatrix} Y \\ Cb \\ Cr \end{pmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.169 & -0.331 & 0.500 \\ 0.500 & -0.419 & -0.081 \end{bmatrix} \begin{pmatrix} R \\ G \\ B \end{pmatrix} + \begin{pmatrix} 0 \\ 128 \\ 128 \end{pmatrix} \tag{10}$$

TABLE I
Quantitative Comparison Results on CT-MRI Fusion

Methods	SCD↑	MS-SSIM↑	$Q^{AB/F}$↑	SD↑	Q_w↑	VIF↑
CSMCA [39]	1.060	0.944	0.599	9.285	0.795	0.317
EMFusion [40]	1.387	0.914	0.512	9.615	0.645	0.416
FusionGAN [20]	0.132	0.723	0.207	8.534	0.414	0.198
IFCNN [15]	1.297	0.868	0.538	9.631	0.563	0.349
U2Fusion [16]	0.840	0.836	0.287	9.097	0.575	0.312
IFT [22]	1.119	0.893	0.457	9.322	0.684	0.378
SwinFusion [1]	**1.566**	**0.954**	**0.631**	**9.651**	**0.798**	**0.497**
MACTFusion (Ours)	**1.715**	**0.959**	**0.660**	**9.685**	**0.809**	**0.507**

1) CT-MRI Fusion: We provide three typical CT-MRI image pairs to intuitively demonstrate the fusion performance. The qualitative results are shown in Fig. 7. There are CT images, MRI images, and the fused images of CSMCA, EMFusion, FusionGAN, IFCNN, U2Fusion, IFT, SwinFusion and MACTFusion (ours). It is hoped that the fusion image can simultaneously retain the dense information of CT and the texture details of MRI. We zoom in on an area in the red box. In Fig. 7, the brightness and clarity of FusionGAN, U2Fusion and IFT are unsatisfactory. There are grey matters blur the texture details. Especially, there are some artifacts and redundant information for FusionGAN. IFCNN can preserve structural details well, while the dense structures are weakened. EMFusion and CSMCA can retain the dense structures while lose some edge details. SwinFusion can achieve a satisfactory fusion effect, but the edges of the fused image are relatively sharp. In contrast, our MACTFusion can retain more texture details while retaining appropriate dense structures. Visually, they look more natural and the contrast is more pronounced.

Quantitative comparison results on six metrics are shown in Table I. It can be found that our MACTFusion achieves optimal results on SCD, MS-SSIM, $Q^{AB/F}$, SD, Q_w and VIF. The optimal MS-SSIM and SCD reflect that our fusion images can obtain higher structure similarity and richer information. The optimal SD and Q_w indicate that MACTFusion transfers more contrast and saliency information. Benefiting from the powerful fine-grained feature extraction capability of GRDB, the optimal $Q^{AB/F}$ indicates our fusion images can obtain more edge information. In addition, higher VIF indicates our fusion images have better visual effects. This is also consistent with the results of qualitative evaluation.

2) PET-MRI Fusion: In Fig. 8, there are three pairs of typical PET and MRI images, and the qualitative results of different models. We can observe that the fused images of FusionGAN have lower contrast and missing texture details. CSMCA and IFCNN can retain the texture information well. However, the darker colour indicates that the functional information is not well preserved. In addition, IFT, U2Fusion and EMFusion can preserve the features of functional images well, but the texture details in the salient regions appear less sharp, as shown in

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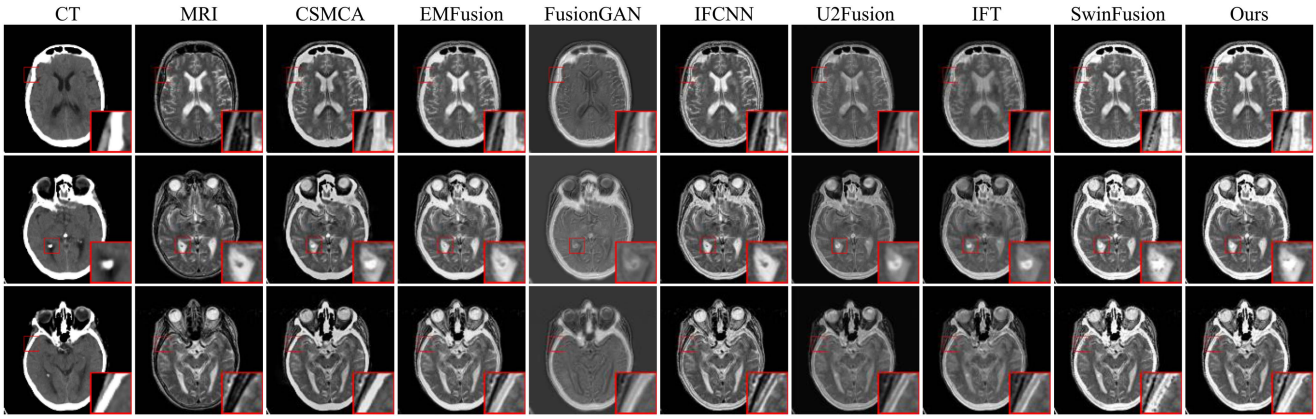


Fig. 7. Qualitative results on fusion of three typical MRI and CT images. They are CT images, MRI images, and fused results of CSMCA, EMFusion, FusionGAN, IFCNN, U2Fusion, IFT, SwinFusion and MACTFusion (ours).

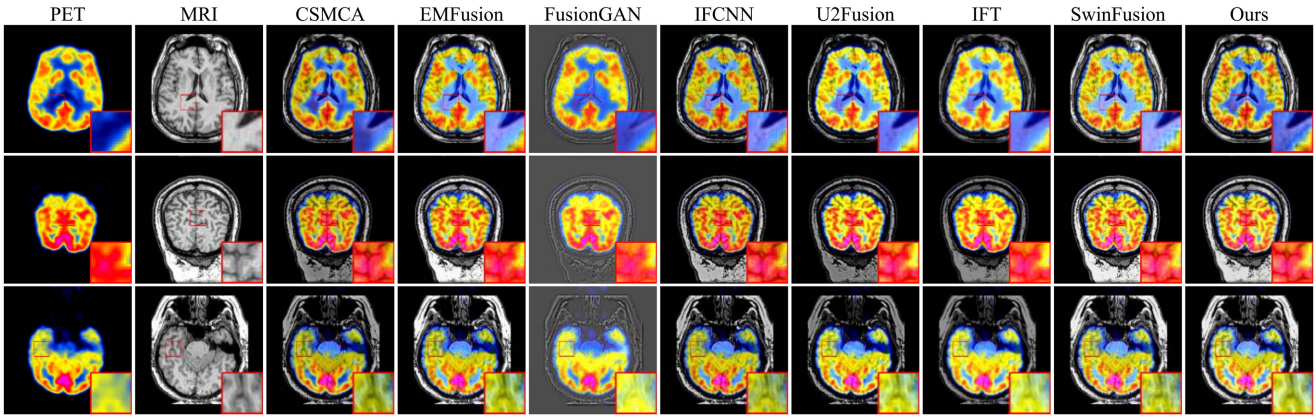


Fig. 8. Qualitative results on fusion of three typical MRI and PET images. They are PET images, MRI images, and fused results of CSMCA, EMFusion, FusionGAN, IFCNN, U2Fusion, IFT, SwinFusion and MACTFusion (ours).

the second row. It is worth noting that the SwinFusion can well preserve texture and structure information, but there is still colour distortion, as shown in the first row. By contrast, the clear edges and texture details are preserved in our fusion results, as shown in the first row. Moreover, as shown in the second and third rows, the color distribution of our fused images is more similar to that of PET images. It alleviates the mosaicism of PET images making them more conform with human visual perception.

Quantitative comparison results on six metrics are shown in Table II. It can be found that our MACTFusion achieves the optimal results on SCD, $Q^{AB/F}$, SD, Q_w and VIF. Our MS-SSIM just follows IFCNN and CSMCA by a narrow margin. The optimal $Q^{AB/F}$ indicates that our fusion image contains more texture and edge information. The high SCD, SD, Q_w and MS-SSIM indicate that MACTFusion retains more structure similarities and meaningful information. Moreover, the optimal VIF reflects that our fusion image has better visual effects.

3) SPECT-MRI Fusion: Qualitative comparison results on SPECT-MRI fusion are illustrated in Fig. 9. They are SPECT images, MRI images, and the fused results of different models.

TABLE II
QUANTITATIVE COMPARISON RESULTS ON PET-MRI FUSION

Methods	SCD↑	MS-SSIM↑	$Q^{AB/F}$↑	SD↑	Q_w↑	VIF↑
CSMCA [39]	1.060	**0.932**	0.507	9.423	0.853	0.507
EMFusion [40]	0.994	0.903	0.736	9.815	0.914	0.455
FusionGAN [20]	0.322	0.628	0.200	7.769	0.458	0.214
IFCNN [15]	1.238	**0.932**	0.689	9.534	0.900	0.520
U2Fusion [16]	1.004	0.892	0.505	8.847	0.733	0.485
IFT [22]	1.300	0.910	0.523	9.278	0.720	0.529
SwinFusion [1]	**1.562**	0.923	**0.743**	**9.877**	**0.922**	**0.553**
MACTFusion (Ours)	**1.691**	0.931	**0.771**	**9.900**	**0.931**	**0.571**

Similar to PET-MRI fusion, our MACTFusion also has advantages. It obtains more details and retains appropriate color information of SPECT images. Specifically, FusionGAN over-retain the functional information resulting in some artifacts. EMFusion and SwinFusion can extract texture information well, while they are insufficient in capturing metabolism information. CSMCA,

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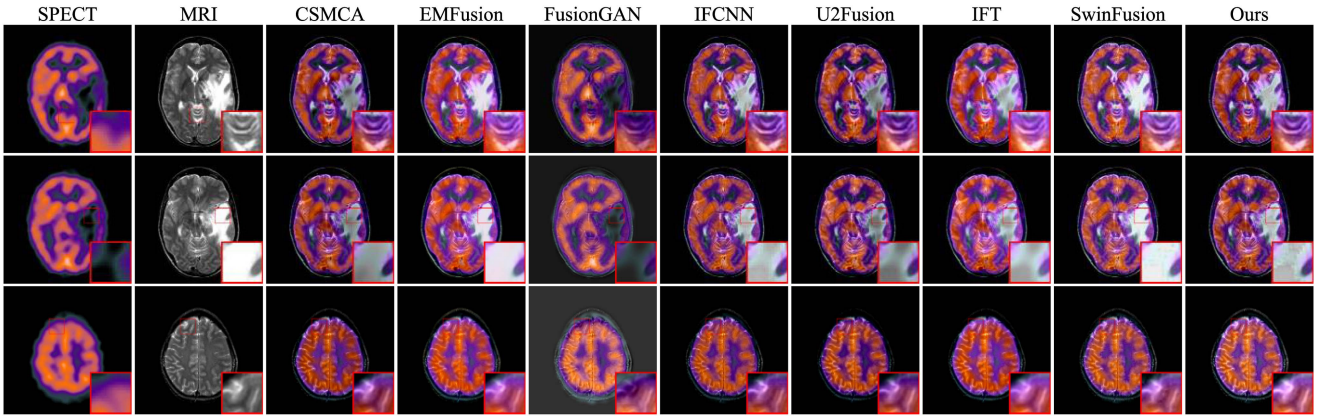


Fig. 9. Qualitative results on fusion of three typical MRI and SPECT images. They are SPECT images, MRI images, and fused results of CSMCA, EMFusion, FusionGAN, IFCNN, U2Fusion, IFT, SwinFusion and MACTFusion (ours).

TABLE III
QUANTITATIVE COMPARISON RESULTS ON SPECT-MRI FUSION

Methods	SCD↑	MS-SSIM↑	$Q^{AB/F}$↑	SD↑	Q_w↑	VIF↑
CSMCA [39]	0.890	0.931	**0.697**	8.124	0.841	0.445
EMFusion [40]	1.082	0.947	0.682	8.287	**0.857**	0.490
FusionGAN [20]	0.190	0.810	0.317	7.431	0.469	0.288
IFCNN [15]	1.220	**0.966**	0.642	8.257	**0.865**	0.549
U2Fusion [16]	0.826	0.937	0.532	7.636	0.798	0.544
IFT [22]	1.217	0.929	0.658	8.274	0.811	0.510
SwinFusion [1]	**1.631**	**0.960**	0.680	**8.393**	0.8511	**0.644**
MACTFusion (Ours)	**1.810**	0.955	**0.699**	**8.415**	0.839	**0.842**

IFCNN, U2Fusion and IFT can also obtain relatively satisfactory fused results, but they cause some loss of fine details.

Quantitative comparison results on six metrics are shown in Table III. It can be observed that our MACTFusion achieves the optimal results on most metrics. The optimal SD and SCD mean that MACTFusion can retain more meaningful information. In addition, the optimal $Q^{AB/F}$ and VIF indicate higher fusion quality and better visual effects. In summary, our MACTFusion can achieve better fusion performance, which contains more functional and morphological information.

C. Clinical Multimodal Brain Tumor Image Fusion

To further validate the effectiveness of MACTFusion, we collaborate with clinicians on clinical multimodal brain tumor image fusion. This study is approved by the Medical Ethics Committee of the First Affiliated Hospital of University of South China (Approval No. 2023KS-FS-07-02). The data are collected from 91 patients consisting of 1368 pairs of images. Among them, 990 are non-pathological images, and 378 are pathological images including brain glioma, metastatic tumor, and meningioma. The image size is 256×256. We increase the images to 10000 through image enhancement. 135 images are randomly selected for testing, with a 2:1 ratio of non-pathological to pathological images. The qualitative results on MR-T1 and MR-T2 fusion in glioma, metastatic tumor and meningioma are illustrated in Fig. 10, respectively. MRI-T1 provides anatomical information on structural abnormalities, such as lesions and tumors. MRI-T2 highlights tissue moisture content and is helpful in detecting edema and inflammation. In Fig. 10, our fused results exhibit higher clarity and richer texture information in both the tumor edge region and brain tissue infiltration. After rigorous evaluations by clinicians, it is found that our fused results can afford a comprehensive insight into the fine characteristics of tumors. This helps doctors evaluate tumor heterogeneity, describe tumor grading characteristics, and further understand the infiltration of tumors into surrounding brain tissue.

D. GFP and PC Image Fusion

To validate the generalization ability of MACTFusion, we conduct experiments on green fluorescent protein (GFP) and phase contrast (PC) image fusion. GFP images provide functional information related to protein distribution, while PC images contain rich details of cell structure including nucleus and mitochondria. GFP-PC fusion can promote biological research, such as gene expression and protein function analysis. We select 22 pairs of GFP and PC images from the public GFP and PC dataset for testing. In Fig. 11, it can be found that CSMCA, FusionGAN, U2Fusion and IFT can well preserve color information, but there is a certain loss of texture details. IFCNN and EMFusion can well retain texture details, but there is slight color distortion. It is worth noting that both SwinFusion and MACTFusion can effectively preserve texture and color information, which proves the effectiveness of cross fusion strategies in image fusion.

E. Application to Medical Image Segmentation

To demonstrate the effectiveness of our MACTFusion in improving downstream tasks, we apply the fusion results to medical image segmentation. We perform segmentation tasks using MR-Flair and MR-T2 image pairs from the Brats public dataset (Brats 2019) [47]. MRI-Flair can enhance the visibility of lesions, such as multiple sclerosis plaques and cortical

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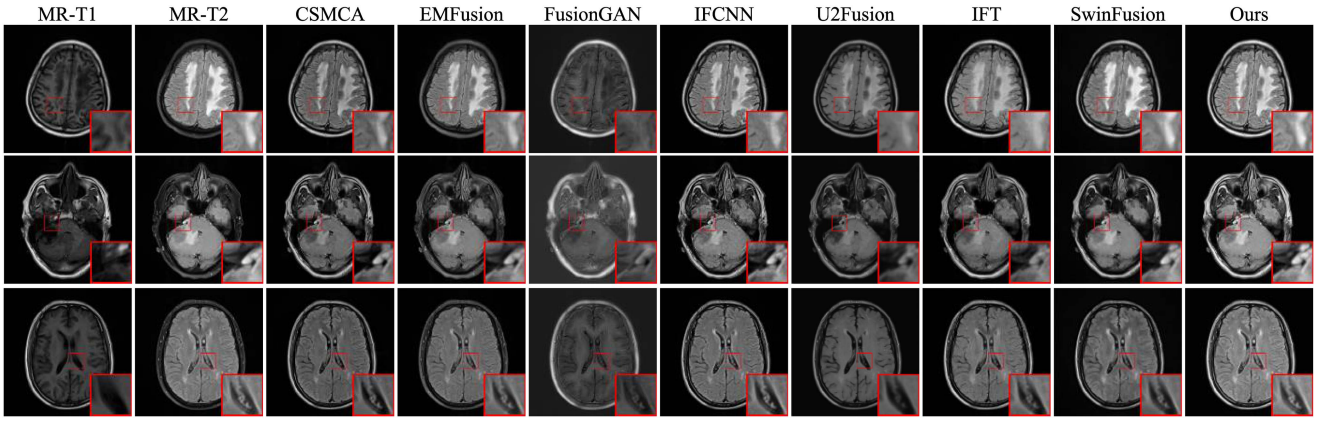


Fig. 10. Qualitative results on fusion of three typical MR-T1 and MR-T2 images. Row 1 is the glioma, row 2 is the brain metastatic tumor, and row 3 is the meningioma.

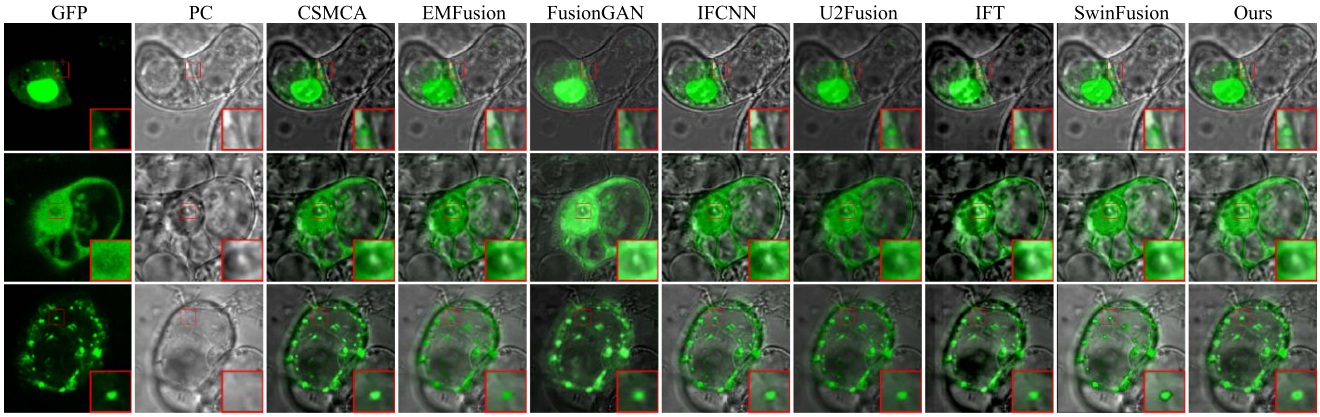


Fig. 11. Qualitative results on fusion of three typical GFP and PC images. They are GFP images, PC images, and fused results of CSMCA, EMFusion, FusionGAN, IFCNN, U2Fusion, IFT, SwinFusion and MACTFusion (ours).

abnormalities. The dataset contains 285 training sets and 49 test sets. The resolution is 240×240×155. We chose Unet [48] as the image segmentation network. In Fig. 12, there are three typical MR-Flair and MR-T2 image pairs, and the fusion results of SwinFusion and MACTFusion, respectively. In the segmentation results, Unet+MR-Flair and Unet+MR-T2 are the segmentation results of single modal MR-Flair and MR-T2, respectively. SwinFusion+Unet are the segmentation results of SwinFusion images, and MACTFusion+Unet are the segmentation results of MACTFusion images. Our MACTFusion segmentation results are more accurate than that of single modal images and SwinFusion images. Since MACTFusion can enhance the details while maintaining the contrast of images, it helps to improve the segmentation performance. Although SwinFusion retains a lot of details, the contrast is slightly imbalanced, resulting in a decrease in segmentation accuracy.

TABLE IV
QUANTITATIVE RESULTS OF THE DESIGNED NETWORK STRUCTURE UNDER DIFFERENT ABLATION EXPERIMENTS

Fusion Tasks		SCD↑	MS-SSIM↑	$Q^{AB/F}$↑	SD↑	Q_w↑	VIF↑
CT-MRI	a	1.661	0.897	0.601	9.667	0.702	0.493
	b	1.691	0.900	0.637	9.648	0.795	0.497
	c	1.683	0.844	0.634	9.650	0.789	0.491
	d	**1.715**	**0.959**	**0.661**	**9.685**	**0.809**	**0.507**
PET-MRI	a	1.601	0.913	0.706	9.878	0.921	0.561
	b	1.680	0.829	0.760	9.844	0.929	0.501
	c	1.613	0.890	0.764	9.852	0.930	0.534
	d	**1.691**	**0.931**	**0.771**	**9.900**	**0.931**	**0.571**
SPECT-MRI	a	1.529	0.945	0.592	8.398	0.820	0.570
	b	1.687	0.937	0.614	8.295	0.830	0.609
	c	1.584	0.922	0.691	8.288	0.823	0.517
	d	**1.810**	**0.955**	**0.699**	**8.415**	**0.839**	**0.842**

F. Ablation Study

1) Analysis of Network Architecture: To demonstrate the effectiveness of designed modules, we conduct extensive ablation experiments on these modules, including shallow feature extraction module (SFE), spatial adaptation fusion module (SAFM), and cross-modality fusion Transformer (CMFT). The quantitative results of the designed network structure under different ablation experiments are reported in Table IV. The quantitative

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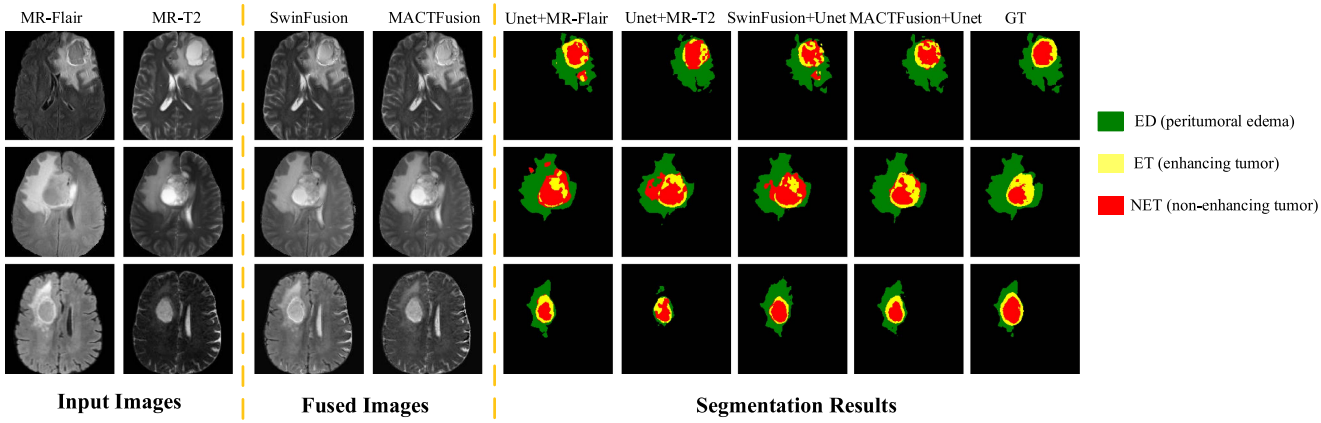


Fig. 12. From left to right: column 1 and column 2 are the MR-Flair and MR-T2 images respectively; column 3 and column 4 are the fusion results with SwinFusion and MACTFusion respectively; column 5 and column 6 are the single modality segmentation results of MR-Flair and MR-T2 with Unet respectively; column 7 and column 8 are the multimodal segmentation results of SwinFuion+Unet and MACTFusion+Unet respectively.

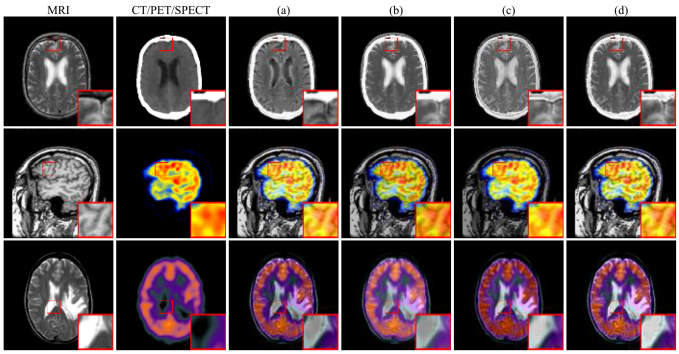


Fig. 13. Qualitative results of the designed network structure under different ablation experiments.

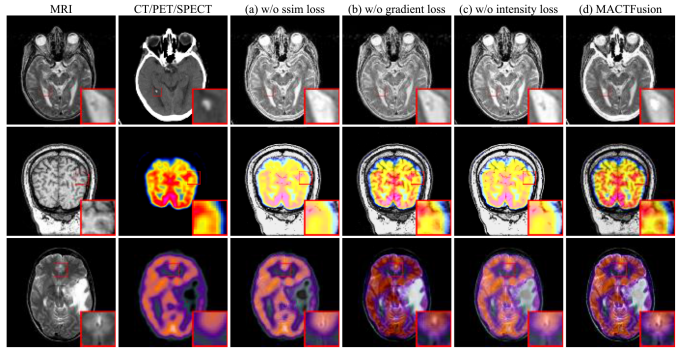


Fig. 14. Qualitative results of the loss function under different ablation experiments.

results of our MACTFusion are represented in (d). We conduct experiment (a) without SFE and find that $Q^{AB/F}$ and Q_w decrease. This indicates that the lack of shallow feature extraction with gradient operations, the ability of network to extract fine-grained details is weakened. Experiment (b) is conducted by replacing SAFM with element addition. We can find that SD and MS-SSIM decrease with respect to our MACTFusion. It reflects that contrast and gray distributions are affected due to the lack of SAFM for remapping feature distributions. Furthermore, we conduct experiment (c) by replacing CMFT with a generic fusion approach, which employs a Transformer lacking the inter-domain cross-operation. We can observe that compared to our MACTFusion, the values of SCD, MS-SSIM and VIF decrease. Since it mainly focuses on intra-domain interactions, the complementary information transferred is limited.

The corresponding qualitative results are shown in Fig. 13. In Fig. 13(a), it can be found that without SFE, the fused images lose some edge details. Especially for MRI-CT fusion, some grey matter blurs texture details. In Fig. 13(b), we can observe that without SAFM, the contrast distribution is unsatisfactory. Especially for MRI-SPECT fusion, it over-retains functional information, resulting in low contrast and loss of texture details. In Fig. 13(c), we can observe that without CMFT, the fusion results are not sufficiently integrated for supplementary information. In contrast, in Fig. 13(d), we can observe that our fusion results can retain sufficient functional information and anatomical details, which further proves the effectiveness of cross-modality fusion Transformer in mining and integrating local and global interactions of multimodal features.

2) Analysis of Loss Function: The defined joint loss function includes structural similarity loss $\mathcal{L}_{ssim}$, gradient loss$\mathcal{L}_{grad}$, and intensity loss $\mathcal{L}_{int}$. We conduct ablation experiments to verify the impact of different sub losses. The qualitative results are shown in Fig. 14. In Fig. 14(a), without loss $\mathcal{L}_{ssim}$, the fused images cannot well retain structure and intensity information. In Fig. 14(b), without loss $\mathcal{L}_{grad}$, the fused images fail to preserve rich texture details. In Fig. 14(c), without loss $\mathcal{L}_{int}$, the fused images cannot maintain appropriate intensity information. Especially, color information is weakened. By contrast, thanks to the joint loss function, our MACTFusion can effectively preserve textures while maintaining appropriate intensity information.

The quantitative results are reported in Table V. It can be observed that removing any sub loss function will result in a decrease in fusion performance. Specifically, without (a) structural similarity loss $\mathcal{L}_{ssim}$, SCD and MS-SSIM significantly decrease. Without (b) gradient loss $\mathcal{L}_{grad}$, the values of $Q^{AB/F}$ decrease.

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TABLE V
QUANTITATIVE RESULTS OF THE LOSS FUNCTION UNDER DIFFERENT ABLATION EXPERIMENTS

Fusion Tasks		SCD↑	MS-SSIM↑	$Q^{AB/F}$↑	SD↑	Q_w↑	VIF↑
CT-MRI	a	1.479	0.778	0.640	9.574	0.719	0.496
	b	1.570	0.948	0.596	9.640	0.718	0.483
	c	1.686	0.943	0.649	9.556	0.612	0.482
	d	**1.715**	**0.959**	**0.661**	**9.685**	**0.809**	**0.507**
PET-MRI	a	1.530	0.758	0.579	9.382	0.673	0.564
	b	1.681	0.870	0.518	8.920	0.838	0.561
	c	1.535	0.835	0.587	9.374	0.702	0.551
	d	**1.691**	**0.931**	**0.771**	**9.900**	**0.931**	**0.571**
SPECT-MRI	a	1.365	0.760	0.626	8.266	0.809	0.526
	b	1.499	0.863	0.553	8.410	0.793	0.585
	c	1.513	0.918	0.630	8.288	0.619	0.617
	d	**1.810**	**0.955**	**0.699**	**8.415**	**0.839**	**0.842**

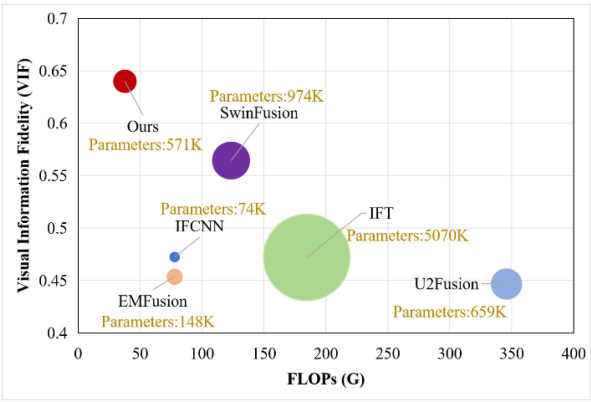


Fig. 15. Comparison results of computational cost and VIF.

Without (c) intensity loss $\mathcal{L}_{\text{int}}$, the values of Q_w are relatively small. These are consistent with qualitative results. By contrast, in the case of joint loss function, our MACTFusion achieves the optimal results on all evaluation metrics.

In addition, in loss function $\mathcal{L} = \delta_1\mathcal{L}_{\text{ssim}} + \delta_2\mathcal{L}_{\text{grad}} + \delta_3\mathcal{L}_{\text{int}}$, there are parameters δ_1, δ_2 and δ_3 that control the trade-off between $\mathcal{L}_{\text{ssim}}$, $\mathcal{L}_{\text{grad}}$ and$\mathcal{L}_{\text{int}}$, respectively. To choose appropriate parameter values, we compare quantitative results on PET-MRI fusion with different δ_1, δ_2 and δ_3 in {1, 10, 20}. We find that, when $\delta_1 = 1$, $\delta_2 = 10$ and $\delta_3 = 1$, better fusion performance is achieved. Therefore, we choose parameter values of δ_1, δ_2 and δ_3 as 1, 10 and 1, respectively. The same values of these parameters are used on the other fusion tasks.

G. Computational Complexity Analysis

The number of training parameters and FLOPs are employed to evaluate the implementation efficiency. We report the results of CNN-based methods including IFCNN, U2Fusion and EMFusion, as well as the Transformer-based methods including IFT and SwinFusion in Fig. 15. We can find that our FLOPs is the smallest, and the number of training parameters is also smaller than those of IFT and SwinFusion. Compared with CNN-based fusion methods, our MACTFusion is also competitive. We also test our running time during the training process, which is about 4 times faster than that of SwinFusion. It is worth noting that the VIF of MACTFusion is the best, hence our method is efficient and effective.

V. CONCLUSION

In this work, we propose a new lightweight adaptive cross Transformer architecture, named MACTFusion, for multimodal medical image fusion. The designed cross multi-axis attention mechanism includes a cross window attention for integrating local features and a cross grid attention for integrating global features from different modalities. Specifically, we introduce a spatial adaption fusion module to guide the cross Transformer. It focuses the model on the most relevant information. In addition, we design a special feature extraction module that combines multiple gradient residual dense convolution and Transformation layers to obtain local features from coarse to fine, and effectively capture global features. It is worth noting that the proposed model minimizes computational costs while maintaining excellent fusion performance. Extensive experiment results have shown that our model outperforms other advanced fusion technologies. Therefore, it has important application value in subsequent diagnosis, treatment planning, and surgical navigation. In the future, we will attempt to apply the proposed MACTFusion to clinical medical image fusion. In addition, we will extend its application to 3D image fusion.

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